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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 14	
TITLE : Exception Handling using Try-Catch	

Problem Statement

Write a Java program to demonstrate exception handling using try and catch blocks , nested and multiple try-catch blocks.

Learning Objectives

To handle runtime exceptions using try-catch , nested try , and multiple catch blocks.

Questions :

1. Create an ATM program handling invalid withdrawal amount using try-catch

2. Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.

Programme

1:

Demo.j ~/Desktop/college_assignments/DSA assignments/demo.c

```
import java.util.*;

class BankAccount
{

        int balance;

    BankAccount(int a)
    {
        this.balance = a;
    }

    void withdrawal(int amount)
    {

        if (amount <= 0)
        {
            throw new IllegalArgumentException("Invalid withdrawal amount");
        }

```

```java

            if (balance < amount)
            {
                System.out.println("Insufficient balance");
            }
            else
            {
                balance -= amount;
                System.out.println("Withdrawal successful. New balance: " + balance);
            }
        }
    }

```

```java
public class Demo
{

    public static void main(String[] args)
    {

        BankAccount ba = new BankAccount(1000);

          Scanner sc = new Scanner(System.in);

        System.out.println("Enter the amount you want to withdraw: ");

         int amount = sc.nextInt();

        try
            {
            ba.withdrawal(amount);
        }
        catch (Exception e)
            {
            System.out.println("Error: " + e);
            }

        sc.close();
    }
}
```

2:

Demo.java

```java
import java.util.*;
class Shopping {
    void order(String product, int quantity) {
        if (quantity <= 0) {
            throw new IllegalArgumentException("Quantity must be greater than zero");
        }
        System.out.println("Product: " + product);
        System.out.println("Quantity: " + quantity);
        System.out.println("Order placed successfully");
    }
}
public class Demo {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter product name: ");
        String product = sc.nextLine();

        System.out.print("Enter quantity: ");
        int quantity = sc.nextInt();

        try {
            Shopping s = new Shopping();
            s.order(product, quantity);
        }
        catch (IllegalArgumentException e) {
            System.out.println("Error: " + e.getMessage());
        }

        sc.close();
    }
}
```

Output

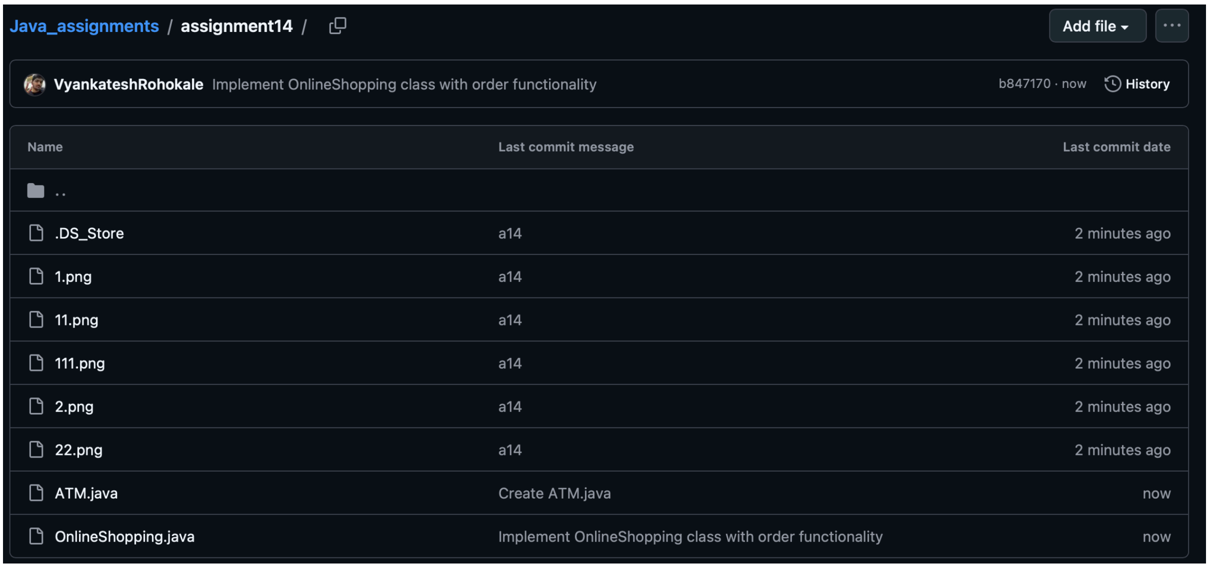
1:

```
piyushrohokale@Piyushs-MacBook-Air DSA assignments % javac Demo.java && java Demo
Enter the amount you want to withdraw:
500
Withdrawal successful. New balance: 500
piyushrohokale@Piyushs-MacBook-Air DSA assignments % javac Demo.java && java Demo
Enter the amount you want to withdraw:
-111
Error: java.lang.IllegalArgumentException: Invalid withdrawal amount
piyushrohokale@Piyushs-MacBook-Air DSA assignments %
```

2:

```
piyushrohokale@Piyushs-MacBook-Air DSA assignments % javac Demo.java && java Demo
Enter product name: apple
Enter quantity: 5
Product: apple
Quantity: 5
Order placed successfully
piyushrohokale@Piyushs-MacBook-Air DSA assignments % javac Demo.java && java Demo
Enter product name: apple
Enter quantity: -1
Error: Quantity must be greater than zero
piyushrohokale@Piyushs-MacBook-Air DSA assignments %
```

Github:



Name	Last commit message	Last commit date
..		
.DS_Store	a14	2 minutes ago
1.png	a14	2 minutes ago
11.png	a14	2 minutes ago
111.png	a14	2 minutes ago
2.png	a14	2 minutes ago
22.png	a14	2 minutes ago
ATM.java	Create ATM.java	now
OnlineShopping.java	Implement OnlineShopping class with order functionality	now

Conclusion:

The programs successfully demonstrate exception handling using try-catch in Java. The ATM program handles invalid withdrawal amounts , while the Online Shopping program handles invalid product quantities. Appropriate error messages are displayed without terminating the program abnormally.